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Student Number:.....

2024 Higher School Certificate Trial Examination Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA-approved calculators may be used
- Write your student number on the paper
- Three data sheets and a Periodic Table are provided at the back of this paper

Total marks: 100

Section I – 20 marks

- Attempt Questions 1-20 (pages 2-10)
- Allow about 35 minutes for this section

Section II – 80 marks

- Attempt Questions 21-31 (pages 11-33)
- Allow about 2 hours and 25 minutes for this section

Disclaimer

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Section I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1-20

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ A. 2 B. 6 C. 8 D. 9

A ☐ B ☒ C ☐ D ☐

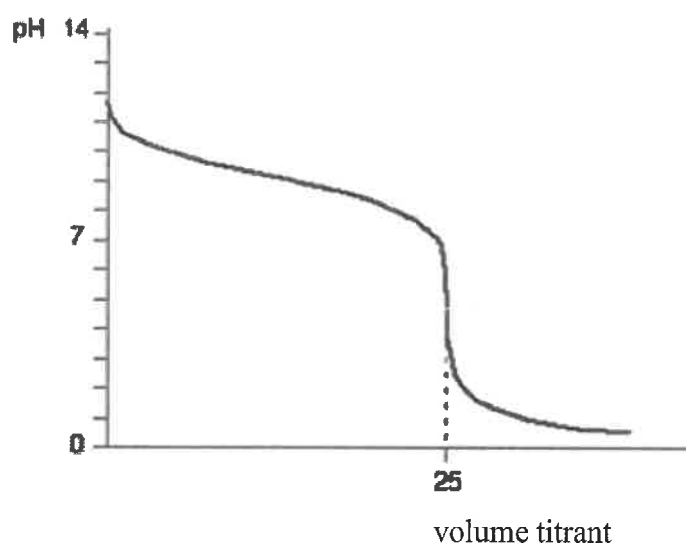
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you have changed your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ ^{*correct*} C ☐ D ☐

- 1** The diagram below represents a particular acid-base titration.



Which one of the following statements about this titration is correct?

- A. Phenolphthalein is a suitable indicator for this reaction.
- B. A solution of a strong base has been delivered by the burette.
- C. The titrant is a solution of a weak acid.
- D. At the equivalence point, the pH of the solution is close to 4.5.

- 2 What is the systematic name of $\text{CH}_3\text{C}(\text{CH}_3)\text{ClCH}_2\text{CH}_3$?
- A. 1-chloro-1,1-dimethylpropane
 - B. 3-chloro-3-methylbutane
 - C. 2-chloro-2-methylbutane
 - D. 3-chloropentane
- 3 Sodium hydroxide solutions carefully prepared from analytical-grade NaOH pellets cannot be used as primary standards in acid-base titrations. This is because
- A. the NaOH solutions prepared are colourless.
 - B. strong bases are caustic and can cause chemical burns.
 - C. the NaOH pellets are too difficult to dissolve when placed into volumetric flasks.
 - D. the exact concentration of the NaOH solutions made from the pellets is unknown.
- 4 Ammonia can be synthesised by reacting nitrogen and hydrogen according to the following equation:
- $$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \Delta H = -92 \text{ kJ mol}^{-1}$$
- The rate at which equilibrium is established can be lowered by increasing the
- A. temperature of the reaction mixture at a constant volume.
 - B. total volume of the reaction vessel at a constant temperature.
 - C. concentration of the reactants at a constant temperature.
 - D. pressure in the reaction vessel at a constant temperature.

- 5 For the reaction pathway shown below, which of the following correctly identifies Reagent X and Product Z?



	Reagent X	Product Z
A.	NaOH	1-propyl ethanoate
B.	$\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$	ethyl propanoate
C.	$\text{H}^+/\text{MnO}_4^-$	1-propyl ethanoate
D.	O_2	ethyl ethanoate

- 6 Which one of the following statements about propene is not correct?
- A. It undergoes an addition reaction with hydrogen to form propane.
 - B. It undergoes a polymerisation reaction, expelling water in the process.
 - C. Weak dispersion forces act between propene molecules and consequently it is a gas at room temperature.
 - D. One propene molecule will react with excess oxygen to produce three molecules of water and three molecules of carbon dioxide.
- 7 15.0 mL of 10.0 mol L⁻¹ HCl is added to 60.0 mL of deionised water. What is the concentration of the diluted acid?
- A. 3.33 mol L⁻¹
 - B. 2.50 mol L⁻¹
 - C. 2.00 mol L⁻¹
 - D. 0.500 mol L⁻¹

- 8 A sealed gas syringe contains an equilibrium mixture of brown nitrogen dioxide gas, NO_2 , and colourless dinitrogen tetroxide gas, N_2O_4 .

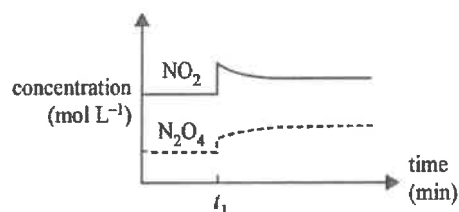
The equation for this equilibrium reaction is:



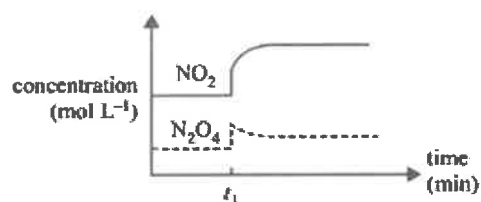
The plunger is suddenly pushed in at time t_1 . The temperature of the gas mixture is kept constant.

Which one of the following concentration–time graphs represents the change that occurs?

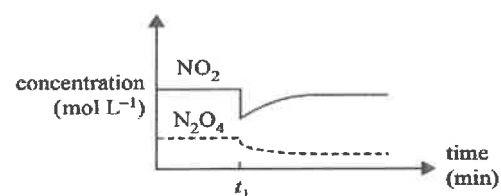
A.



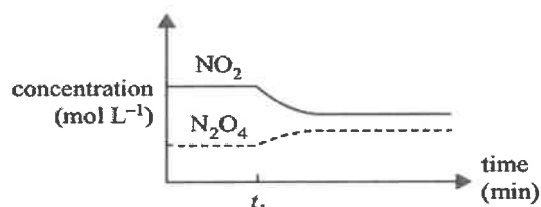
B.



C.



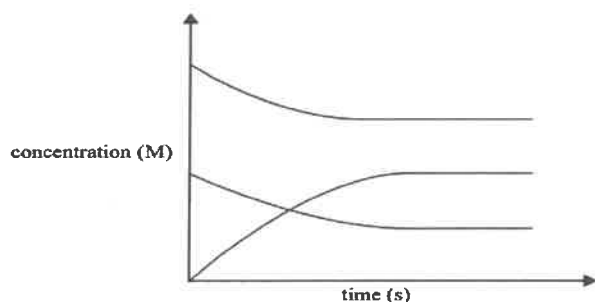
D.



- 9 A condensation reaction involving 200 glucose molecules, $\text{C}_6\text{H}_{12}\text{O}_6$, results in a polysaccharide. What is the molar mass, in g mol^{-1} , of the polysaccharide?
- A. 36000
B. 35982
C. 32418
D. 32400
- 10 The ammonium ion NH_4^+ acts as a weak acid according to the equation:
$$\text{NH}_4^+ (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{NH}_3 (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq}) \quad K_a = 5.6 \times 10^{-10}$$

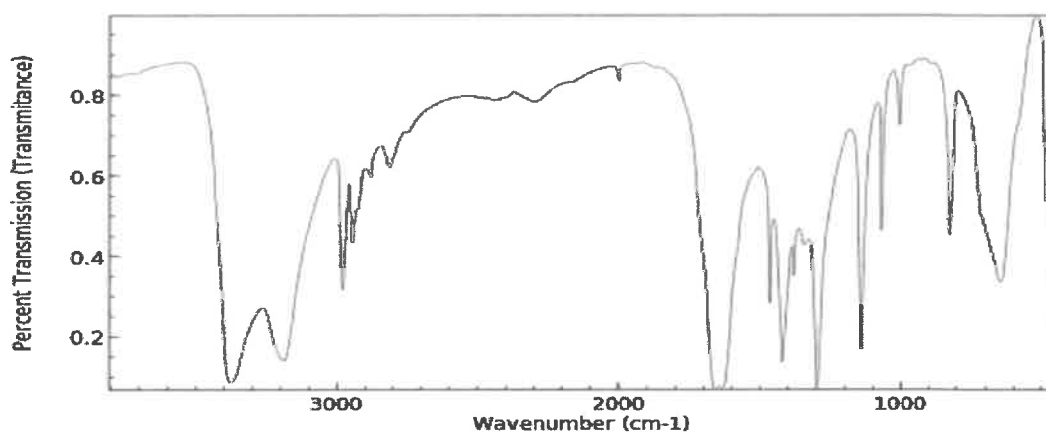
The $[\text{H}_3\text{O}^+]$ of a 0.200 mol L^{-1} ammonium chloride solution is closest to:
- A. $4.79 \times 10^{-6} \text{ mol L}^{-1}$
B. $9.55 \times 10^{-6} \text{ mol L}^{-1}$
C. $1.06 \times 10^{-5} \text{ mol L}^{-1}$
D. $1.51 \times 10^{-5} \text{ mol L}^{-1}$
- 11 What is the name of the product formed when chlorine, Cl_2 , reacts with but-1-ene?
- A. 1,2-dichlorobutane
B. 1,4-dichlorobutane
C. 2,2-dichlorobutane
D. 2,3-dichlorobutane

- 12 Consider the following concentration versus time graph:



Which one of the following reactions could produce the graph shown above?

- A. $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$
- B. $\text{H}_2(g) + \text{Br}_2(g) \rightleftharpoons 2\text{HBr}(g)$
- C. $2\text{HCl}(g) \rightleftharpoons \text{H}_2(g) + \text{Cl}_2(g)$
- D. $2\text{SO}_3(g) \rightleftharpoons 2\text{SO}_2(g) + \text{O}_2(g)$
- 13 An organic compound Z does not react as an acid or a base. The solid dissolves readily in water at 25°C, forming strong hydrogen bonds. The infra-red spectrum of compound Z is shown below:



NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)

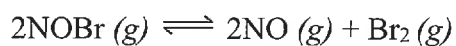
The compound could be

- A. $\text{CH}_3\text{CH}_2\text{COOH}$
- B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- C. $\text{CH}_3\text{CH}_2\text{CONH}_2$
- D. $\text{NH}_2\text{CH}_2\text{COOH}$

14 Which molecule can exist as a *cis* and *trans* isomer?

- A. Hex-1-ene
- B. Hex-2-ene
- C. 2-methylpent-1-ene
- D. 2-methylbut-2-ene

15 Nitrosyl bromide decomposes according to the following equation:



A 0.32 mol sample of NOBr is placed in an evacuated 1.00 L flask. After the system comes to equilibrium, the flask contains 0.23 mol NOBr.

What are the concentrations of NO and Br₂ in the flask at equilibrium?

	[NO] mol L ⁻¹	[Br ₂] mol L ⁻¹
A.	0.09	0.045
B.	0.09	0.09
C.	0.18	0.09
D.	0.46	0.23

- 16 A 2.0 g sample of silver chloride ($MM = 143.4 \text{ g mol}^{-1}$) was added to 100.0 mL of water in a beaker.

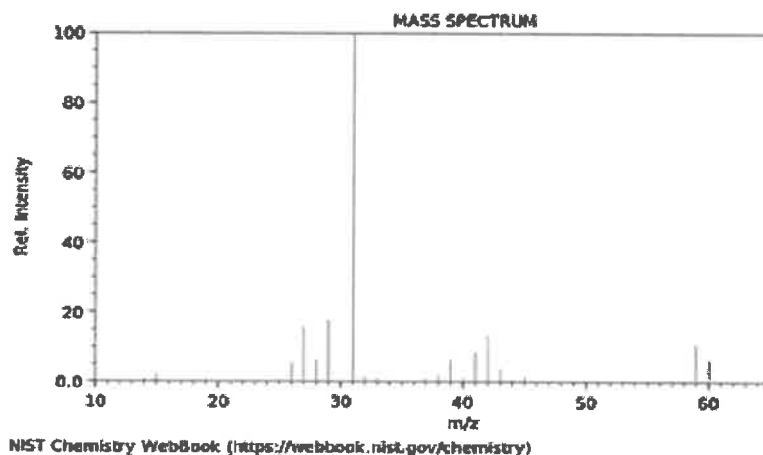
The solubility of silver chloride at this temperature is $1.77 \times 10^{-10} \text{ mol L}^{-1}$.

The sample was then diluted by adding another 100.0 mL of water.

What is the ratio of the concentration of silver ions in solution before and after dilution?

- A. 1:1
 - B. 1:2
 - C. 2:1
 - D. 4:1
- 17 Which of the following is an Arrhenius base?
- A. Na
 - B. $\text{Ba}(\text{OH})_2$
 - C. Na_2CO_3
 - D. HPO_4^{2-}
- 18 Which pair of ions can be distinguished by addition of chloride ions?
- A. Na^+ and K^+
 - B. Ba^{2+} and Ca^{2+}
 - C. Mg^{2+} and Ca^{2+}
 - D. Pb^{2+} and Zn^{2+}

- 19 An unidentified organic substance is found to react with an acid to form an ester. The mass spectrum of this compound is shown below.



Which one of the following species is consistent with the base peak on the mass spectrum?

- A. $[\text{CH}_2\text{OH}]^+$
 - B. $[\text{CH}_3\text{CH}_2\text{O}]^+$
 - C. $[\text{CH}_3\text{O}]^+$
 - D. $[\text{COOH}]^+$
- 20 The ^{13}C NMR spectrum of an organic compound has two peaks. The ^1H NMR spectrum of the same compound has only 1 peak, with no splitting. The compound is
- A. propanal.
 - B. propanoic acid.
 - C. propane.
 - D. propanone.

Student Number:

Chemistry

Section II – 80 marks

Attempt Questions 21-31

Allow about 2 hours and 25 minutes for this section

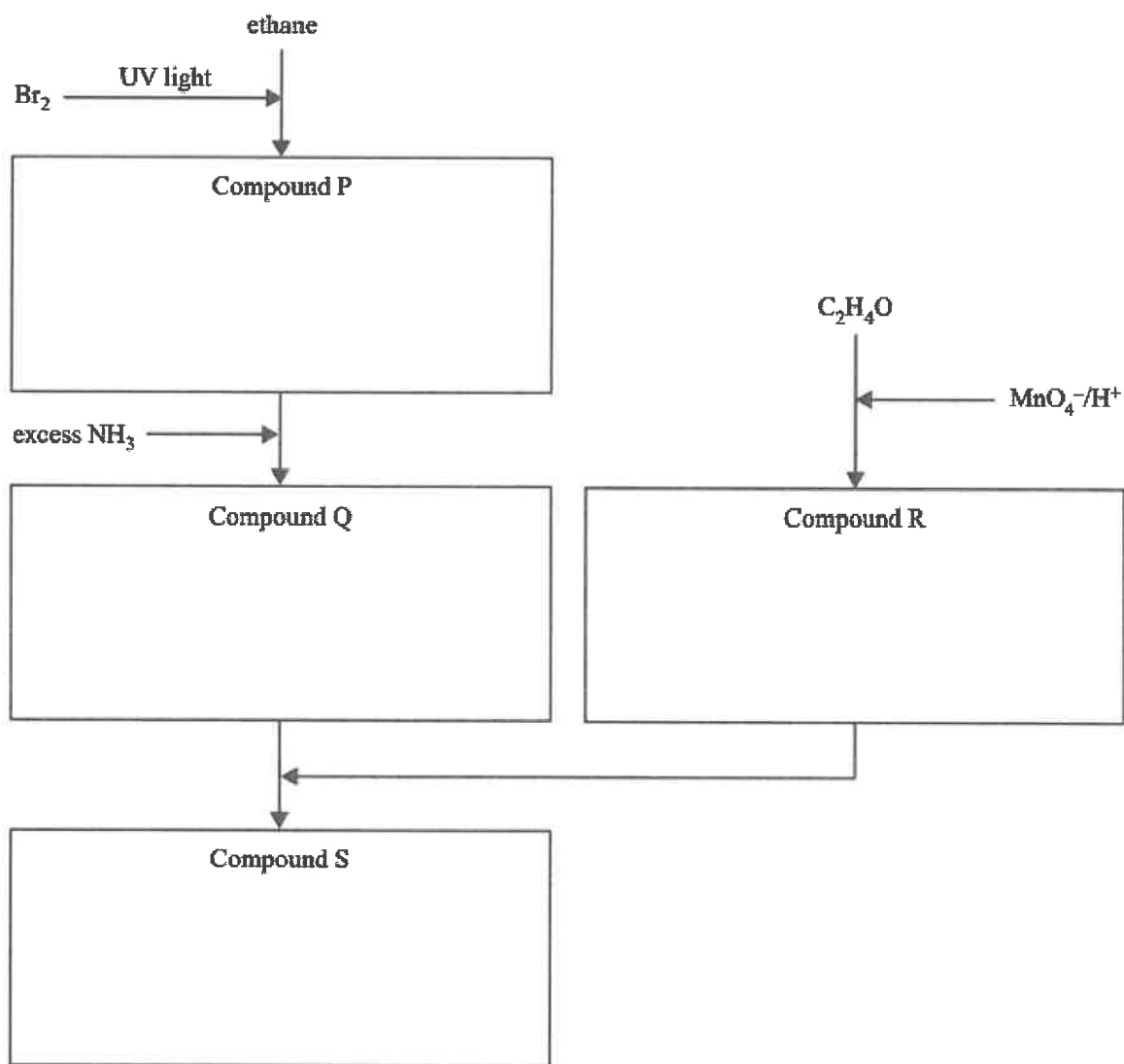
Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided on pages 34 and 35. If you use this space, clearly indicate which question you are answering.

Question 21 (8 marks)

The diagram below shows a reaction pathway involving carbon compounds, starting with ethane and $\text{C}_2\text{H}_4\text{O}$. Assume that the ethane and bromine react in a 1:1 mole ratio and that only 1 compound containing carbon atoms is formed.



- (a) Write the IUPAC name of Compound P in the box provided. 1
- (b) Name the type of reaction that produces Compound Q from Compound P. 1
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- (c) Write the structural formula of Compound Q in the box provided. 1

Question 21 continues on the next page

Question 21 (continued)

- (d) When $\text{C}_2\text{H}_4\text{O}$ is mixed with potassium permanganate, KMnO_4 , and sulfuric acid, H_2SO_4 , a carboxylic acid, Compound R, is produced. 2

Draw the structure of Compound R in the box provided and name Compound R below.

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- (e) What is the function of the H_2SO_4 in the reaction to produce Compound R? 1

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- (f) Compound Q reacts with Compound R in a condensation reaction to produce Compound S. 2

Draw the structural formula for Compound S in the box provided and name Compound S below.

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Question 22 (4 marks)

A scientist used quantitative analysis of an unknown compound **A**, which contains only carbon, oxygen, and chlorine, to determine its molecular formula.

- (a) A 3.00 g sample of **A** was completely vaporised in a 1.00 L container at 70.0°C and exerted a pressure of 86.51 kPa. What is the molar mass of **A**? 1

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- (b) A 0.300 g sample of **A** was added to 100 mL water, which resulted in the conversion of all the chlorine in **A** to HCl. After briefly bubbling nitrogen gas through the solution, the hydrochloric acid was titrated with 0.2000 mol L⁻¹ NaOH solution. 2

The titration required 30.33 mL of added NaOH to reach a phenolphthalein endpoint. What is the mass percentage of chlorine in compound **A**?

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- (c) What is the molecular formula for **A**? 1

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Question 23 (9 marks)

The following table lists the pH of 0.10 mol L^{-1} solutions of four different acids at 25°C .

Acid	pH
I	1.0
II	3.0
III	0.7
IV	2.1

- (a) Which one of the four acids listed in the table above has the smallest K_a value? 1

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- (b) Which acid must have more than one acidic hydrogen per molecule? Give a reason for your answer. 1

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- (c) Using the concentration and the pH of acid IV, calculate the percentage ionisation of acid IV in the 0.10 mol L^{-1} solution. 1

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Question 23 continues on the next page

Question 23 (continued)

- (d) Samples of the solutions of acids I and IV are diluted by a factor of 10. For which of these acids is the change in pH greater? Explain your answer.

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- (e) Methanoic acid is a weak monoprotic acid, with $K_a = 1.8 \times 10^{-4}$. Calculate the concentration of a methanoic acid solution that will have the same pH as acid IV.

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- (f) The dissociation of methanoic acid in water is exothermic. If a solution of the acid is heated, will the pH of the solution increase, decrease or remain constant? Give an explanation for your answer.

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Question 24 (6 marks)

Polymers are significant organic molecules. They are classified on the basis of the method of linking of their monomers. Choose TWO polymers you have studied which have different methods of linkage.

For each of these polymers:

- Name the polymer
- Classify the polymer according to its type of linkage
- Draw a monomer or monomers
- Draw part of a polymer chain that includes at least 4 monomer units

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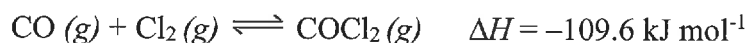
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Question 25 (8 marks)

Phosgene, COCl_2 , is a highly toxic gas used in chemical industry. COCl_2 is usually produced by the reaction between carbon monoxide and chlorine according to the equation.



A scientist conducted two experiments on the equilibrium reaction of COCl_2 .

The experiments were conducted in evacuated and sealed 4 L containers at 150°C .

Experiment 1: 2 mol of $\text{COCl}_2(\text{g})$ was injected into a previously evacuated, sealed 4 L container.

Experiment 2: 4 mol of $\text{COCl}_2(\text{g})$ was injected into another previously evacuated, sealed 4 L container.

(a) Which experiment had the higher initial rate of production of Cl_2 ? Justify your answer. 2

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- (b) If, for Experiment 1, the concentrations of $\text{COCl}_2(g)$ and $\text{CO}(g)$ were equal at equilibrium, $[\text{COCl}_2] = [\text{CO}]$, then what conclusion could be made about the relative concentrations of COCl_2 and CO in Experiment 2 at equilibrium? Justify your answer. 3

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Question 25 continues on the next page

Question 25 (continued)

- (c) 2 mol of an inert gas is injected into the container in Experiment 1. The temperature is kept at 150°C. What effect will this have on the rate of production of the Cl_2 in the container? Justify your answer using collision theory.

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- (d) The temperature for Experiment 2 is increased to 200°C. Explain the effect on the equilibrium concentration of COCl_2 in the reaction.

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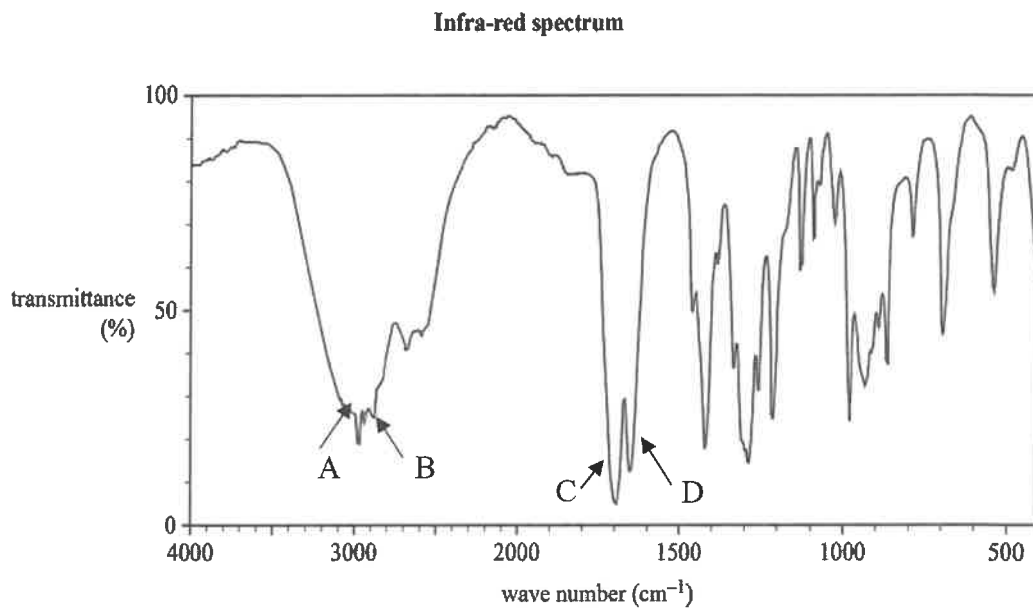
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Question 26 (13 marks)

A student investigated an organic substance, Compound Y, with the molecular formula $C_5H_8O_2$.

- (a) The infra-red spectrum of Compound Y is shown below. On the spectrum, FOUR peaks have been labelled.

3



Data: SDBSWeb, <<http://sdbs.db.aist.go.jp>>,
National Institute of Advanced Industrial Science and Technology

Identify the bonds responsible for each of the 4 peaks identified on the spectrum.

A =

B =

C =

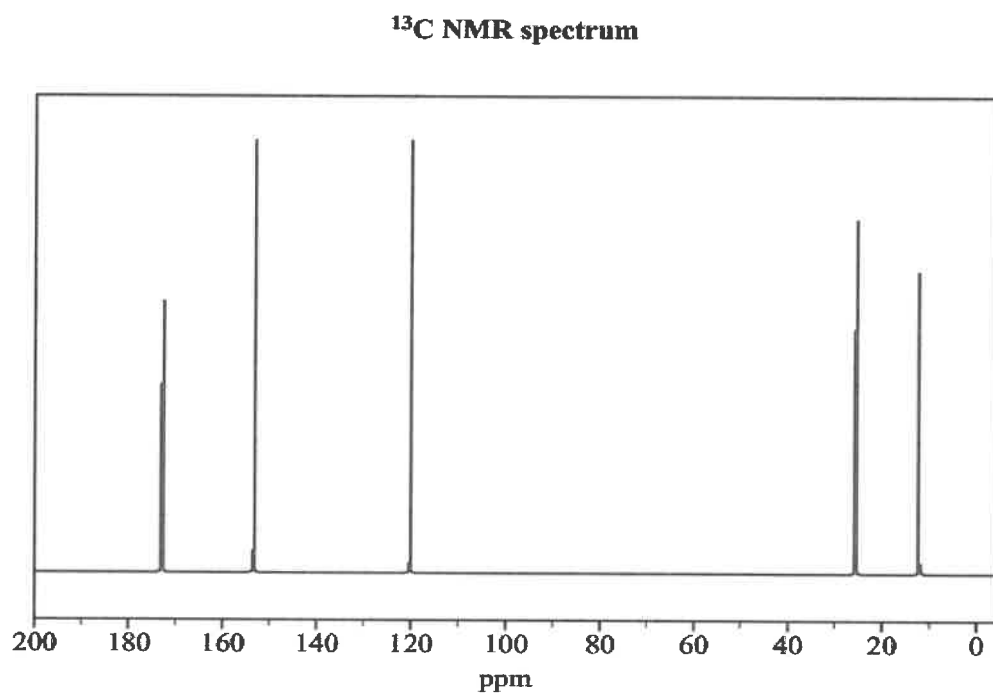
D =

Question 26 continues on the next page

Question 26 (continued)

- (b) A sample of Compound Y was further analysed using ^{13}C NMR spectroscopy. The spectrum is shown below.

2



Data: SDBSWeb, <<http://sdb.s.db.aist.go.jp>>,
National Institute of Advanced Industrial Science and Technology

Use the information provided in the ^{13}C NMR spectrum to identify the number of different carbon environments for Compound Y and explain the significance of the difference in the position of the peaks on the spectrum.

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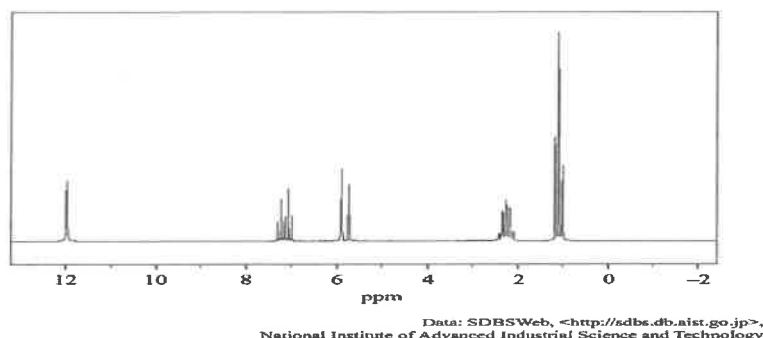
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Question 26 continues on the next page

Question 26 (continued)

- (c) A sample of Compound Y was further analysed using ^1H NMR spectroscopy. The spectrum is shown below.

2



^1H NMR data

Chemical shift (ppm)	Relative peak area	Peak splitting
1.1	3	triplet (3)
2.3	2	pentet (5)
5.8	1	doublet (2)
7.1	1	quartet (4)
12	1	singlet (1)

For the signal at 1.1 ppm in the ^1H NMR spectrum, identify what specific information is provided by the relative peak area and the peak splitting.

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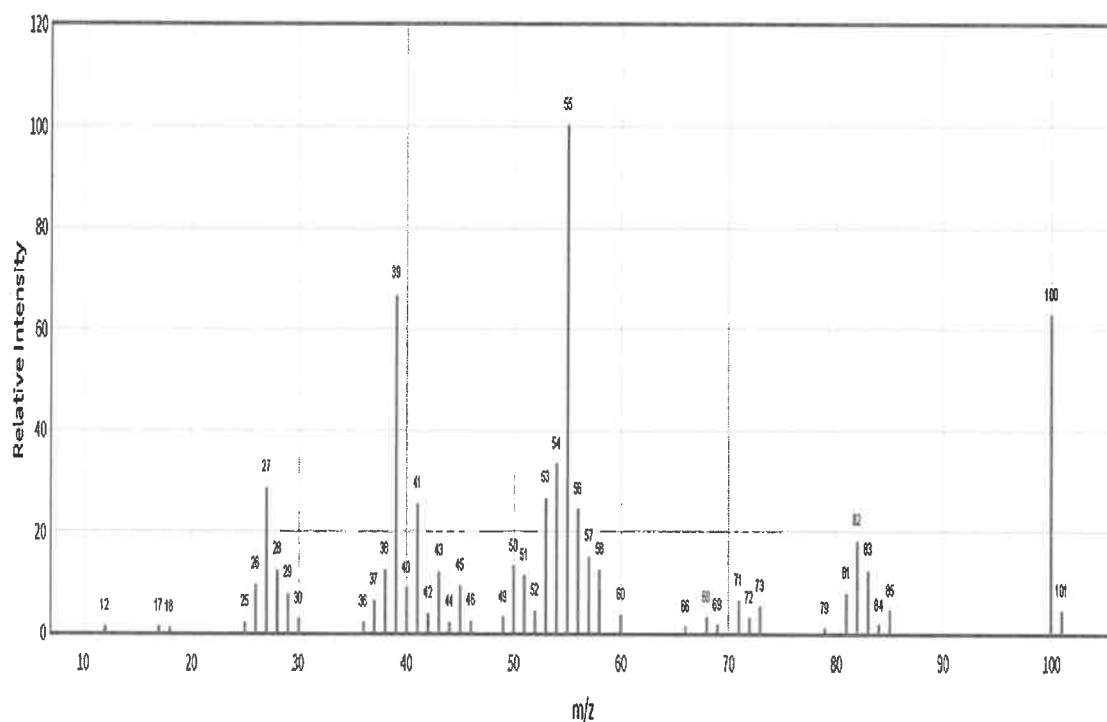
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Question 26 continues on the next page

Question 26 (continued)

(d) The mass spectrum of Compound Y is shown below.

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Identify the parent peak and the base peak and suggest the structure of the fragment formed which has the m/z value shown at the base peak.

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Question 26 continues on the next page

Question 26 (continued)

- (e) Draw the structural formula of Compound Y. Justify your answer by relating aspects of the structure to the evidence from the spectra and data table. 4

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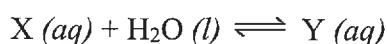
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Question 27 (6 marks)

When substance X is dissolved in water, it reacts to form an equilibrium mixture with substance Y according to the equation:



The concentration of X can be determined using UV-visible spectroscopy. X absorbs strongly at 290 nm. Y shows no absorption at this wavelength.

In a particular experimental arrangement at 25°C, the relationship between absorbance at 290 nm and concentration of X in mol L⁻¹ is given by:

$$\text{Absorbance} = 3.75 \times [\text{X}]$$

In the experiment, 0.110 mol of X is dissolved rapidly in 1.00 L of water at 25°C. The absorbance of the solution changes as some of the X is converted to Y. The table below shows the change in absorbance over time (measured in seconds).

Absorbance	0.389	0.274	0.244	0.230	0.226	0.226
Time (s)	6.00	60.0	90.0	120	240	480

- (a) Calculate the concentration of X, in mol L⁻¹, when the reaction reached equilibrium. 1

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- (b) Calculate the absorbance at the instant that X was dissolved in the water before any reaction occurred. 1

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- (c) Calculate the percentage of the original 0.110 mol of X that has been converted into Y at equilibrium. 2

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Question 27 continues on the next page

Question 27 (continued)

- (d) The average rate of a reaction can be determined by calculating the change in concentration of a reactant per second. Calculate the average rate, in $\text{mol L}^{-1} \text{s}^{-1}$, at which the concentration of X changed during the first 6.00 s of the reaction. 2

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Question 28 (5 marks)

- (a) Calculate the mass of lead(II) fluoride ($K_{sp} = 4.0 \times 10^{-8}$) which will dissolve in 1.00 L of water. **2**

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- (b) Determine the molar solubility of lead(II) fluoride in 1.00 L of solution to which 0.10 mol of solid lead(II) nitrate has been added. **2**

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- (c) The solubility of lead(II) fluoride increases with temperature. **1**

Write an equation for the equilibrium reaction of lead(II) fluoride in water and include information about the enthalpy change for this reaction.

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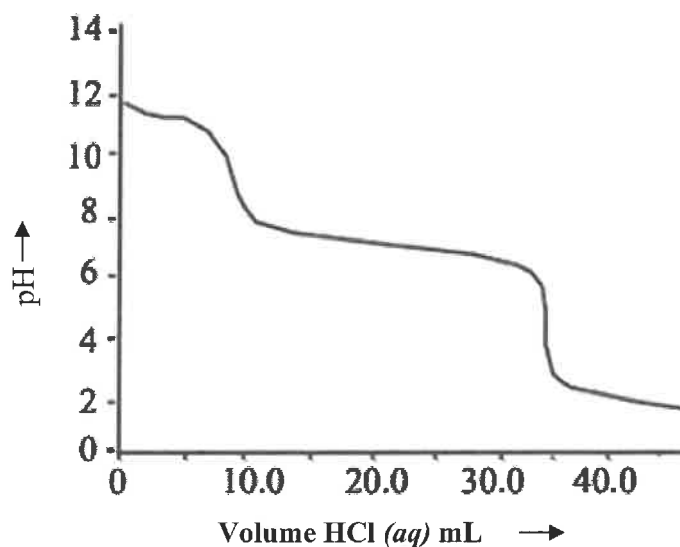
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Question 29 (10 marks)

The percentages of NaHCO_3 and Na_2CO_3 are to be determined in a mixture of them with KCl.

A 0.500 g sample of the mixture is dissolved in 50.0 mL of deionised water and titrated with $0.115 \text{ mol L}^{-1} \text{ HCl}$. A student produced this sketch of the pH titration curve.



- (a) Write balanced equations for the reaction that is responsible for the equivalence point that occurs at about pH = 9 and pH = 5. 2

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- (b) Calculate the total number of moles of acid used to reach each equivalence point if the volumes are 9.63 mL and 34.27 mL, respectively. 2

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Question 29 continues on the next page

Question 29 (continued)

- (c) Determine the number of grams of Na_2CO_3 , NaHCO_3 and KCl and their percentages in the original mixture. 4

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- (d) Sketch a titration curve for a solution of Na_2CO_3 by itself and describe how it differs from the given curve. 2

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Question 30 (6 marks)

Consider the reaction shown in the following equation:



- (a) Write an expression for the equilibrium constant for this reaction.

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- (b) 10.0 mol of NOBr, 10.0 mol of NO and 5.0 mol of Br₂ are placed in a 1.0 L container at 1000 K. Predict in which direction the reaction will proceed. Justify your answer.

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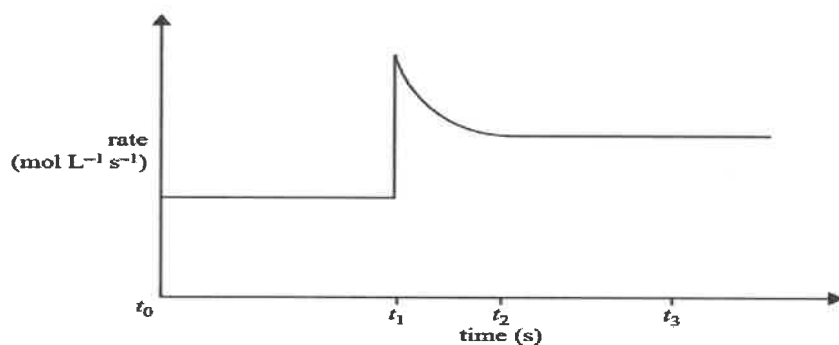
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Question 30 continues on the next page

Question 30 (continued)

- (c) A mixture of NO, NOBr and Br₂ is initially at equilibrium. The following graph shows how the rate of formation of NOBr in the mixture changes when the volume of the reaction vessel is decreased at time t_1 . 3



Use collision theory and factors that affect the rate of a reaction to explain the shape of the graph at the time intervals indicated.

- (i) between t_0 and t_1

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- (ii) at t_1

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- (iii) between t_1 and t_2

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Question 31 (5 marks)

- (a) Write an equation for the complete combustion of octane, C_8H_{18} . 1

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- (b) Write an equation for the fermentation of glucose to produce ethanol. 1

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- (c) What are “biofuels”? Outline 2 benefits and 2 problems associated with the use of a named biofuel. 3

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End of Examination

Section II - Extra writing space

If you use this space, clearly indicate which question you are answering.

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Section II - Extra writing space

If you use this space, clearly indicate which question you are answering.

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Student Number:

CHEMISTRY – MULTIPLE-CHOICE ANSWER SHEET

ATTEMPT ALL QUESTIONS

Question	1	A ☐	B ☐	C ☐	D ☐
	2	A ☐	B ☐	C ☐	D ☐
	3	A ☐	B ☐	C ☐	D ☐
	4	A ☐	B ☐	C ☐	D ☐
	5	A ☐	B ☐	C ☐	D ☐
	6	A ☐	B ☐	C ☐	D ☐
	7	A ☐	B ☐	C ☐	D ☐
	8	A ☐	B ☐	C ☐	D ☐
	9	A ☐	B ☐	C ☐	D ☐
	10	A ☐	B ☐	C ☐	D ☐
	11	A ☐	B ☐	C ☐	D ☐
	12	A ☐	B ☐	C ☐	D ☐
	13	A ☐	B ☐	C ☐	D ☐
	14	A ☐	B ☐	C ☐	D ☐
	15	A ☐	B ☐	C ☐	D ☐
	16	A ☐	B ☐	C ☐	D ☐
	17	A ☐	B ☐	C ☐	D ☐
	18	A ☐	B ☐	C ☐	D ☐
	19	A ☐	B ☐	C ☐	D ☐
	20	A ☐	B ☐	C ☐	D ☐

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Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_0}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Infrared absorption data

Bond	Wavenumber/cm ⁻¹
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C—O	1000–1300
C—C	750–1100

¹³C NMR chemical shift data

Type of carbon	δ/ppm
$\begin{array}{c} \quad \\ -C-C- \\ \quad \end{array}$	5–40
$\begin{array}{c} \\ R-C-Cl \text{ or } Br \\ \end{array}$	10–70
$\begin{array}{c} \\ R-C-C- \\ \quad \\ O \end{array}$	20–50
$\begin{array}{c} \\ R-C-N \\ \quad \diagup \quad \diagdown \end{array}$	25–60
$\begin{array}{c} \\ -C-O- \\ \end{array}$ alcohols, ethers or esters	50–90
$\begin{array}{c} \diagdown \quad \diagup \\ C=C \\ \diagup \quad \diagdown \end{array}$	90–150
R—C≡N	110–125
	110–160
$\begin{array}{c} R-C- \\ \\ O \end{array}$ esters or acids	160–185
$\begin{array}{c} R-C- \\ \\ O \end{array}$ aldehydes or ketones	190–220

UV absorption

(This is not a definitive list and is approximate.)

Chromophore	λ _{max} (nm)
C—H	122
C—C	135
C=C	162

Chromophore	λ _{max} (nm)
C≡C	173 178 196 222
C—Cl	173
C—Br	208

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY																2 He 4.003 Helium																	
		Atomic Number Symbol Standard Atomic Weight Name																																	
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		79 Au 197.0 Gold														5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine									
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium																13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon							
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids 89–103		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po Polonium		85 At Astatine		86 Rn Radon	
87 Fr Francium		88 Ra Radium		Actinoids		104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium		111 Rg Roentgenium		112 Cn Copernicium		113 Nh Nihonium		114 Fl Flerovium		115 Mc Moscovium		116 Lv Livermorium		117 Ts Tennessee		118 Og Oganesson	



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Lanthanoids

Actinoids

Standard atomic weights are abridged to four significant figures. Elements with no reported values in the table have no stable nuclides. Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

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2024 HIGHER SCHOOL CERTIFICATE CHEMISTRY EXAMINATION MARKING GUIDELINES & ADVICE FOR TEACHERS

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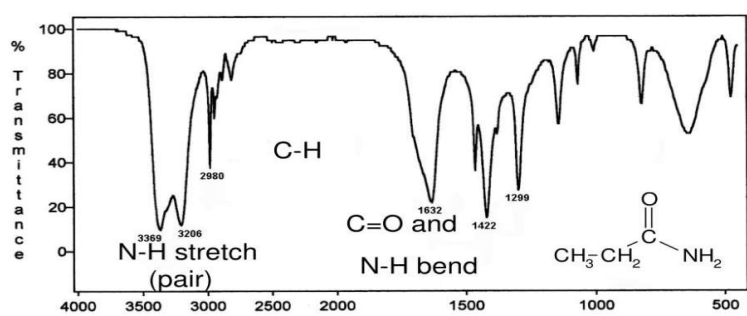
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2024 Higher School Certificate Trial Examination Chemistry

Marking Guidelines Section I

Question	Answer	Explanation to assist students and teachers
1	D	The titration is a strong acid against a weak base. The weak base is in the flask; the strong acid is in the burette. The equivalence point is at 4.5, so methyl orange, not phenolphthalein, should be used.
2	C	The chain is numbered to give the -methyl and -chloro groups the lower number. Hence 2-chloro-2-methylbutane.
3	D	Sodium hydroxide pellets can absorb water and react with carbon dioxide as they are being weighed out. Hence NaOH is not used as a primary standard as the exact concentration is unknown.
4	B	Increasing the total volume of the container decreases the concentration of the gases, so the number of collisions per second is reduced and hence the rate of reaction to reach equilibrium is lowered. The other 3 options (A, C and D) would all increase the rate of reaction.
5	C	Reagent X must be an oxidising agent (acidified permanganate ion). The ester formed is 1-propyl ethanoate.
6	B	Alkenes undergo addition reactions, not condensation reactions which release water, to form polymers.
7	C	Mol HCl present = $n = (15.0/1000) \times 10.0 = 0.150 \text{ mol}$ Volume after dilution = $75.0 \text{ mL} = 0.075 \text{ L}$ $c = n/V = 0.150/0.075 = 2.00 \text{ mol L}^{-1}$
8	A	When the plunger is pushed in, there is a spike in the concentration of both gases. The equilibrium will shift to the side with fewer particles to compensate for the applied change. Hence the spike will be followed by an increase in $[\text{N}_2\text{O}_4]$ and a decrease in $[\text{NO}_2]$.

Question	Answer	Explanation to assist students and teachers
9	C	When 200 glucose molecules react to form a polysaccharide by condensation polymerisation, 199 water molecules are released. $MM(\text{polysaccharide}) = (200 \times m(\text{C}_6\text{H}_{12}\text{O}_6)) - (199 \times m(\text{H}_2\text{O}))$ $= (200 \times 180.0) - (199 \times 18.0)$ $= 32418 \text{ g mol}^{-1}$
10	C	$K_a = \frac{[\text{NH}_3][\text{H}_3\text{O}^+]}{[\text{NH}_4^+]}$ $[\text{NH}_4^+] = 0.200 \text{ mol L}^{-1} \text{ and } [\text{NH}_3] = [\text{H}_3\text{O}^+]$ $K_a = [\text{H}_3\text{O}^+]^2 / 0.200$ $K_a(\text{NH}_4^+) = 5.6 \times 10^{-10} = [\text{H}_3\text{O}^+]^2 / 0.200$ $[\text{H}_3\text{O}^+]^2 = 0.200 \times 5.6 \times 10^{-10}$ $[\text{H}_3\text{O}^+] = \sqrt{(1.12 \times 10^{-10})} = 1.06 \times 10^{-5} \text{ mol L}^{-1}$
11	A	The Cl ₂ molecule adds across the double bond, so is between atoms 1 and 2 in the 4-carbon chain.
12	B	The change in concentration of the 2 reactants is identical. The change in the concentration of the product is twice that of the change in reactants. This would not be the case in reaction A – where the reactants react in a 1:3 ratio. In reaction C and D, 1 reactant produces two products. From the graph, 2 reactants are producing 1 product, so B is the correct answer.
13	C	The reactant Z is an amide. Low molecular mass amides are solids and dissolve readily in water. The answer must be C propanamide, as all the other molecules listed can act as an acid or base or as both. The IR spectrum is not characteristic of an acid or alcohol in the 3500-3000 cm⁻¹ range but does have evidence of a carbonyl group at 1680 cm⁻¹.  The IR spectrum shows transmittance (%) on the y-axis (0 to 100) and wavenumber (cm⁻¹) on the x-axis (4000 to 500). Key features include: a broad N-H stretch (pair) around 3300 cm⁻¹ (labeled 3369 and 3206); C-H stretches around 2900 cm⁻¹ (labeled 2980); a strong C=O carbonyl stretch at 1632 cm⁻¹; an N-H bend at 1422 cm⁻¹; and other peaks at 1299 and 1422 cm⁻¹. The chemical structure of propanamide, CH₃-CH₂-C(=O)-NH₂, is shown.

Question	Answer	Explanation to assist students and teachers															
14	B	$\begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_2 - \text{CH}_3 \\ \diagdown \quad / \\ \text{C} = \text{C} \\ / \quad \diagdown \\ \text{H} \quad \text{H} \end{array}$ Cis – isomer $\begin{array}{c} \text{H}_3\text{C} \quad \text{H} \\ \diagdown \quad / \\ \text{C} = \text{C} \\ / \quad \diagdown \\ \text{H} \quad \text{CH}_2 - \text{CH}_3 \end{array}$ Trans – isomer The isomers which have the same structural formula but differ in the spatial arrangement of the groups around the double bond are called geometrical isomers. This geometrical isomerism is shown by alkenes or their derivatives. In the <i>cis</i>-isomer, similar groups reside on the same side, while the similar groups reside on opposite sides in a <i>trans</i> isomer. A and C must be incorrect as the double bond is between atoms 1 and 2 and there are 2 hydrogen atoms on carbon-1. D is incorrect as there are 2 methyl groups attached to carbon-2.															
15	A	$(0.32 - 0.23) = 0.090$ moles NOBr₂ reacts to reach equilibrium. <table style="margin-left: auto; margin-right: auto;"><tr><td>$2\text{NOBr} (g)$</td><td>$\rightleftharpoons$</td><td>$2\text{NO} (g)$</td><td>+</td><td>$\text{Br}_2 (g)$</td></tr><tr><td>2 mol</td><td></td><td>2 mol</td><td></td><td>1 mol</td></tr><tr><td>0.090 mol</td><td></td><td>0.090 mol</td><td></td><td>0.045 mol</td></tr></table> Hence 0.09 mol NO and 0.045 mol Br₂ form in the 1 L flask.	$2\text{NOBr} (g)$	$\rightleftharpoons$	$2\text{NO} (g)$	+	$\text{Br}_2 (g)$	2 mol		2 mol		1 mol	0.090 mol		0.090 mol		0.045 mol
$2\text{NOBr} (g)$	$\rightleftharpoons$	$2\text{NO} (g)$	+	$\text{Br}_2 (g)$													
2 mol		2 mol		1 mol													
0.090 mol		0.090 mol		0.045 mol													
16	A	The solubility of silver chloride is 1.77×10^{-10} mol/L. Only 1.77×10^{-11} mol silver chloride can dissolve in 100 mL. The number of moles of AgCl added to 100 mL water = $2.0/143.4 = 1.39 \times 10^{-3}$ mol. If an additional 100 mL water is added, an additional 1.77×10^{-11} mol AgCl will dissolve but the volume is now twice the original. So the solution is still saturated, and the concentration of silver ions will remain as previously.															
17	B	An Arrhenius base is defined as a soluble hydroxide, ionising to hydroxide ions in solution. (Note: Substances such as ammonia and the carbonate ion are Brønsted-Lowry bases, but the Arrhenius Theory restricted the definition to hydroxides and excluded those substances that reacted with water to form hydroxide ions.)															
18	D	$\text{Pb}^{2+} (aq) + 2\text{Cl}^- (aq) \rightarrow \text{PbCl}_2 (s)$ $\text{Zn}^{2+} (aq)$ does not form a precipitate with $\text{Cl}^- (aq)$.															
19	A	The compound must be an alcohol. The parent ion is at $m/z = 60$, so it is an isomer of propanol. The base peak is at $m/z = 31$. This corresponds to a particle with 1 carbon, 1 oxygen and 3 hydrogens. A $[\text{CH}_2\text{OH}]^+$ particle would be formed during ionisation. A $[\text{CH}_3\text{O}]^+$ is not possible from a 3 carbon alcohol. This particle would be more likely formed from an ether.															

Question	Answer	Explanation to assist students and teachers
20	D	Propanone has 2 different carbon environments (2 identical -CH_3 groups and a C=O group). Hence 2 peaks on the ^{13}C NMR spectrum. It has only 1 peak on the ^1H NMR, as all 6 hydrogen atoms are in an identical environment. Propanal and propanoic acid have 3 C environments and 3 H environments. Propane has 2 C environments and 2 H environments.

Section II – 80 marks**Question 21** (8 marks)

21 (a) (1 mark)

Criteria	Mark
• Bromoethane (in box)	1

Sample answer

Bromoethane

21 (b) (1 mark)

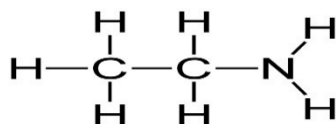
Criteria	Mark
• Substitution	1

Sample answer

Substitution

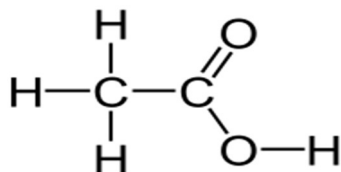
21 (c) (1 mark)

Criteria	Mark
• Correct structural formula (in box)	1

Sample answer

21 (d) (2 marks)

Criteria	Marks
• Correct structural formula of R (in box) AND correct name	2
• Correct structural formula of R (in box) OR correct name	1

Sample answer

Ethanoic acid

21 (e) (1 mark)

Criteria	Mark
• Identifies H_2SO_4 as a reagent/part of oxidising agent (not as a catalyst)	1

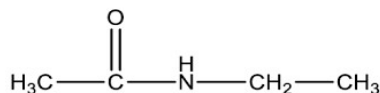
Sample answer

Sulfuric acid is part of the oxidising agent. The combination of MnO_4^- and H^+ gains electrons from the reactant, ethanal.

21 (f) (2 marks)

Criteria	Marks
Draws the structural formula for N-ethylethanamide (in box) AND names the compound as N-ethylethanamide	2
Draws the structural formula for N-ethylethanamide (in box) OR names the compound as N-ethylethanamide	1

Sample answer



N-ethylethanamide

Question 22 (4 marks)

22 (a) (1 mark)

Criteria	Mark
Correctly determines the molar mass of A	1

Sample answer

$$n = PV/RT = (86.51)(1.00 \text{ L})/(8.314 \text{ J mol}^{-1}\text{K}^{-1})(343.2 \text{ K}) = 0.0303 \text{ mol}$$

Hence molar mass of A is $3.00 \text{ g}/0.0303 \text{ mol} = 99.0 \text{ g/mol}$

22 (b) (2 marks)

Criteria	Marks
Correctly determines the mass percentage of chlorine in A	2
Correctly determines the moles of HCl formed	1

Sample answer

Titration results show moles of NaOH used in reaction and hence moles of HCl formed.

$$0.03033 \text{ L NaOH} \times 0.2000 \text{ mol/L} = 0.006066 \text{ mol NaOH}$$

Hence 0.006066 mol HCl formed and 0.006066 mol Cl in the compound

$$\text{Hence mass of Cl in sample} = \text{mol Cl} \times 35.45 \text{ g/mol} = 0.2150 \text{ g Cl in the sample}$$

Hence the mass percentage of chlorine in compound A

$$= (0.2150 \text{ g} / 0.300 \text{ g sample}) \times 100\% = 71.7\% \text{ Cl by mass}$$

22 (c) (1 mark)

Criteria	Mark
COCl_2	1

Sample answer

$$\text{In } 99.0 \text{ g (1 mol) A, there is } (0.717 \times 99.0 \text{ g})/(35.45 \text{ g mol}^{-1} \text{ Cl}) = 2 \text{ mol Cl}$$

$$\text{The remaining mass in 1 mol A is } 99.0 \text{ g/mol} - (2 \times 35.45 \text{ g/mol}) = 28.1 \text{ g/mol}$$

The only combination of atomic masses of C and O that is close to 28 g/mol is C_1O_1

So the molecular formula is COCl_2 .

Question 23 (9 marks)

23 (a) (1 mark)

Criteria	Mark
• Acid II (the acid with pH = 3.0)	1

Sample answer

The acid with the highest pH produces the lowest concentration of H^+ from a 0.10 mol L^{-1} solution. This acid must have the smallest K_a value. So Acid II.

23 (b) (1 mark)

Criteria	Mark
• Correct explanation of why Acid III must have more than one acidic hydrogen	1

Sample answer

The lowest possible pH of a 0.10 mol L^{-1} solution of a monoprotic acid is 1.00. Hence there must be more than 1 hydrogen per molecule in Acid III.

23 (c) (1 mark)

Criteria	Mark
• Correct answer	1

Sample answer

$$\% \text{ ionisation} = ([H^+] / [\text{acid}]) \times 100 = (10^{-2.1} / 0.10) \times 100 = 7.9\%$$

23 (d) (2 marks)

Criteria	Marks
• Explains clearly why the change in pH is greater for Acid I than for Acid IV by calculation of the changes in pH for both acids OR by relating the change to Le Chatelier's Principle	2
• Recognises that the stronger acid will have a greater change in pH	1

Sample answer

Acid I is a strong acid, which ionises completely. A dilution by a factor of 10 will increase the pH by 1 unit, from 1 to 2, as the $[H^+]$ is now reduced by a factor of 10.

Acid IV is a weak acid, so an equilibrium exists in solution.

$$\text{For acid IV, the equilibrium constant} = K_a = (10^{-2.1})^2 / 0.1 = 6.3 \times 10^{-4}$$

When the acid is diluted by a factor of 10, $[HA] = 0.01 \text{ mol L}^{-1}$

$$6.3 \times 10^{-4} = [H^+]^2 / 0.01$$

$$\text{Hence } [H^+] = 0.0025 \text{ mol L}^{-1}$$

$$\text{pH} = 2.6$$

The change in pH for Acid IV is from 2.1 to 2.6, which is less than the pH change than for Acid I.

23 (e) (2 marks)

Criteria	Marks
• Correct answer, with working	2
• Correct method and reasoning, with mathematical error	1

Sample answer

$$K_a(\text{HCOOH}) = 1.8 \times 10^{-4}$$

Since $\text{pH} = 2.1$, the $[\text{H}^+] = 10^{-2.1} = (7.9 \times 10^{-3})$

Hence, $1.8 \times 10^{-4} = (10^{-2.1})^2 / [\text{HCOOH}]$

$$[\text{HCOOH}] = (10^{-2.1})^2 / 1.8 \times 10^{-4} = 0.35 \text{ mol L}^{-1}$$

23 (f) (2 marks)

Criteria	Marks
• Explains that the pH of the solution will increase, because the equilibrium will move to favour the endothermic reverse reaction and the $[\text{H}_3\text{O}^+]$ decreases	2
• States that the $[\text{H}_3\text{O}^+]$ decreases or that the equilibrium shifts to the left to favour the endothermic reaction	1

Sample answer



If the solution is heated, the equilibrium will shift to the left, to compensate for the increase in temperature. An increase in temperature always favours the endothermic reaction in an equilibrium mixture. This means the $[\text{H}_3\text{O}^+]$ will decrease. Hence the pH will increase.

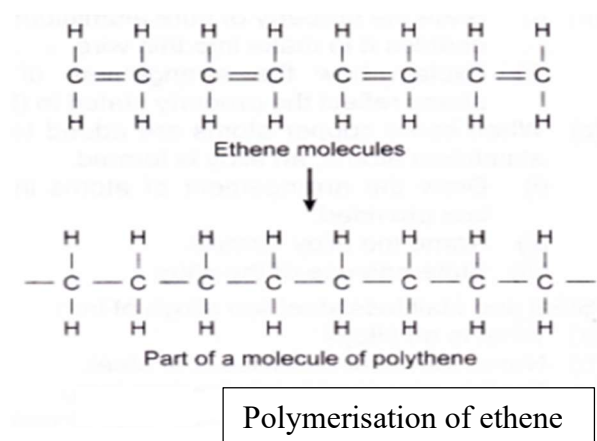
Question 24 (6 marks)

Criteria	Marks
- Correctly classifies ONE named addition polymer and ONE named condensation polymer - For both polymers, correctly draws a monomer - For both polymers, correctly draws part of a polymer chain that includes 4 monomer units	6
- Correctly classifies ONE named addition polymer and ONE named condensation polymer - For both polymers, correctly draws a monomer - For ONE polymer, correctly draws part of a polymer chain that includes 4 monomer units	5
- Correctly classifies ONE named addition polymer AND ONE named condensation polymer - For ONE polymer, correctly draws a monomer and names the polymer - For ONE polymer, correctly draws part of a polymer chain that includes 4 monomer units	4
- Correctly classifies ONE named addition polymer OR ONE named condensation polymer - For the correctly classified polymer, correctly draws a monomer and names the polymer - For the correctly classified polymer, correctly draws part of a polymer chain that includes 4 monomer units	3
- Correctly classifies ONE named addition polymer OR ONE named condensation polymer - For the correctly classified polymer, correctly draws a monomer and names the polymer OR - For the correctly classified polymer, correctly draws part of a polymer chain that includes 4 monomer units	2
Correctly classifies ONE named addition polymer OR ONE named condensation polymer	1

Sample answer (many possible answers)Addition polymer

Polyethylene (polythene)

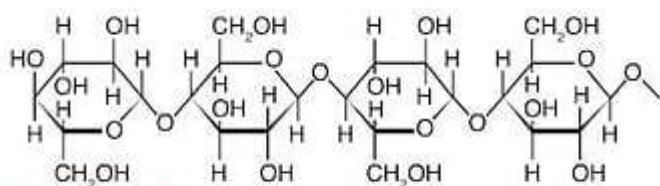
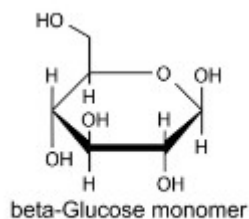
Monomer: ethene



Condensation polymer

Cellulose

Monomer: β -glucose



Cellulose

Question 25 (8 marks)

25 (a) (2 marks)

Criteria	Marks
• Identifies Experiment 2 as having the higher initial rate of production of Cl_2 AND	2
• Justifies the answer in terms of the increased concentration of COCl_2	
• Identifies Experiment 2 as having the higher initial rate of production of Cl_2	1

Sample answer

Chlorine is formed by the reverse reaction in the equilibrium equation. Experiment 2 has a higher initial concentration of COCl_2 molecules, so there will be a higher frequency of collisions (higher number of collisions per unit time) and so a higher frequency of successful collisions producing a faster rate of reaction/rate of production of Cl_2 .

25 (b) (3 marks)

Criteria	Marks
- Identifies $[\text{COCl}_2] > [\text{CO}]$ in Experiment 2 AND - Includes valid reasoning and/or working	3
- Recognises that the 2 reactions have the same K_{eq} AND - Writes correct expressions for the TWO equilibrium constants	2
- Recognises that the 2 reactions have the same K_{eq} OR - Writes correct expressions for the TWO equilibrium constants	1

Sample answer

Since both experiments are at the same temperature and have the same reactants and product, then both will have the same equilibrium constant.

However, the $[\text{Cl}_2]_{\text{Exp.2}}$ is greater than $[\text{Cl}_2]_{\text{Exp.1}}$ (from Part (a) above)

The constant for Experiment 1 = $K_1 = \frac{[\text{COCl}_2]_{\text{Exp.1}}}{[\text{CO}]_{\text{Exp.1}}[\text{Cl}_2]_{\text{Exp.1}}}$
 Since in Experiment 1, $[\text{COCl}_2] = [\text{CO}]$, then $K_1 = 1/[\text{Cl}_2]_{\text{Exp.1}}$

In Experiment 2, $K_2 = \frac{[\text{COCl}_2]_{\text{Exp.2}}}{[\text{CO}]_{\text{Exp.2}}[\text{Cl}_2]_{\text{Exp.2}}}$

But since $K_1 = K_2$

$$\frac{[\text{COCl}_2]_{\text{Exp.1}}}{[\text{CO}]_{\text{Exp.1}}[\text{Cl}_2]_{\text{Exp.1}}} = \frac{1}{[\text{Cl}_2]_{\text{Exp.1}}} = \frac{[\text{COCl}_2]_{\text{Exp.2}}}{[\text{CO}]_{\text{Exp.2}}[\text{Cl}_2]_{\text{Exp.2}}}$$

Since $[\text{Cl}_2]_{\text{Exp.2}}$ is greater than $[\text{Cl}_2]_{\text{Exp.1}}$ (from Part (a) above)

$$\text{Rearranging } \frac{[\text{Cl}_2]_{\text{Exp.2}}}{[\text{Cl}_2]_{\text{Exp.1}}} = \frac{[\text{COCl}_2]_{\text{Exp.2}}}{[\text{CO}]_{\text{Exp.2}}} > 1$$

Then $[\text{COCl}_2]_{\text{Exp.2}} > [\text{CO}]_{\text{Exp.2}}$

A valid answer, in terms of Le Chatelier's principle, including reasoning could also be accepted.

25 (c) (2 marks)

Criteria	Marks
Explains that the rate of production of Cl_2 will not change AND	2
Uses collision theory to explain the answer	1

Sample answer

There will not be any change in the rate of production of Cl_2 . There will be more collisions due to added particles but not more successful collisions since collisions involving inert gas particles do not affect equilibrium. Inert gas molecules are not part of the equilibrium reaction.

25 (d) (1 mark)

Criteria	Mark
Explains that concentration of COCl_2 will decrease as the increase in temperature favours the backward reaction	1

Sample answer

$[\text{COCl}_2]$ will decrease as the increase in temperature favours the backward reaction.

Question 26 (13 marks)

26 (a) (3 marks)

Criteria	Marks
FOUR correct answers.	3
TWO or THREE correct answers.	2
ONE correct answer.	1

Sample answer

The bonds responsible are:

A = O-H (in an acid)

B = C-H

C = C=O

D = C=C

26 (b) (2 marks)

Criteria	Marks
Identifies 5 different carbon environments AND	2
Indicates that 2 of the 5 environments are in non-polar groups and 3 carbons are closer to electronegative atoms	1

Sample answer

There are 5 different carbon environments, as shown by the 5 peaks on the spectrum.

3 carbon atoms are closer to electronegative atoms, in general, are more to the left of the standard.

2 of the 5 environments are in non-polar groups and 3 carbons are closer to electronegative atoms.

26 (c) (2 marks)

Criteria	Marks
- Identifies that the relative area indicates that there are 3 equivalent hydrogen atoms at that chemical shift AND - Identifies that the splitting indicates that there are 2 equivalent hydrogen atoms on adjacent carbon atoms	2
- Indicates that there are 3 equivalent hydrogen atoms at that chemical shift OR - Indicates that there are 2 equivalent hydrogen atoms on adjacent carbon atoms	1

Sample answer

The relative area indicates that there are 3 equivalent hydrogen atoms at that chemical shift.
The splitting pattern indicates that there are 2 equivalent hydrogen atoms on adjacent carbon atoms.

26 (d) (2 marks)

Criteria	Marks
- Identifies the parent peak at 100 and the base peak at 55 AND - Identifies a likely fragment of $m/z = 55$ as $[C_4H_7]^+$	2
Identifies the parent peak at 100 and the base peak at 55	1

Sample answer

The parent peak is at $m/z = 100$

The base peak is at $m/z = 55$

Hence the fragment formed most frequently will have m/z value of 55

The fragment with $m/z = 55$ would correspond to $[C_4H_7]^+$

26 (e) (4 marks)

Criteria	Marks
• Draws the correct structural formula for pentenoic acid • Identifies that the molecular formula indicates a double bond in the compound • Identifies that the ^{13}C NMR spectrum indicates 5 different carbon environments • Identifies that the mass spectrum supports a 5 carbon acid and 1 double bond • Identifies that the IR spectrum supports the presence of $-\text{COOH}$, $\text{C}=\text{O}$ and $\text{C}=\text{C}$ functional groups • Identifies that the ^1H NMR spectrum indicates a CH_3CH_2- group	4
• Draws a partly correct molecule and justifies the structure by evidence from all 4 of the spectra	3
• Draws a partly correct molecule and justifies the structure by evidence from 3 of the spectra	2
• Outlines some correct evidence from the spectra or data provided	1

Sample answer

The molecular formula ($\text{C}_5\text{H}_8\text{O}_2$) indicates that the molecule is unsaturated (a saturated 5 carbon acid or ester would have formula $\text{C}_5\text{H}_{10}\text{O}_2$).

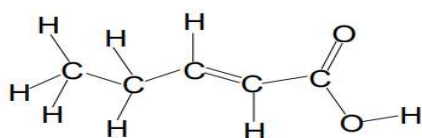
The IR spectrum gives evidence to support an acid, with $\text{C}=\text{O}$ group as part of the $-\text{COOH}$ group and a $\text{C}=\text{C}$ double bond.

The ^{13}C NMR spectrum indicates 5 different carbon environments.

The mass spectrum supports a 5-carbon acid and $-\text{COOH}$ group, as the parent ion corresponds to the molecular mass of 100 and the fragment at the base peak (55) corresponds to a $[\text{C}_4\text{H}_7]^+$ group. This indicates that the parent ion has lost a $-\text{COOH}$ group which then splits into smaller fragments.

The ^1H NMR spectrum (1 carbon with 3 hydrogens and with 2 hydrogens on adjacent carbon atom) supports the presence of an ethyl group (CH_3CH_2-). This must be terminal in the molecule.

The double bond must be between carbon atoms 2 and 3, to create the splitting patterns for the 5 groups of hydrogen atoms.



Question 27 (6 marks)

27 (a) (1 mark)

Criteria	Mark
• Correct answer	1

Sample answer

Equilibrium occurs at 240 s

Absorbance at equilibrium = 3.75 [X]_{equil} = 0.226[X]_{equil} = 0.226/3.75 = 0.0603 mol L⁻¹

27 (b) (1 mark)

Criteria	Mark
• Correct answer	1

Sample answer[X] = 0.110 mol L⁻¹

Absorbance initially = 3.75 x 0.110 = 0.4125

27 (c) (2 marks)

Criteria	Marks
• Correct answer	2
• Correct method, with mathematical error	1

Sample answer

Change in %X at equilibrium = ((0.110 – 0.0603)/0.110) x 100 = 45.2%

27 (d) (2 marks)

Criteria	Marks
• Correct answer (0.001 mol L ⁻¹ s ⁻¹)	2
• Calculates the [X] at 6.00 s	1

Sample answer

Absorbance at 6.00 s = 0.389

[X] at 6.00 s = 0.389/3.75 = 0.104 mol L⁻¹ $\Delta[X] = 0.110 - 0.104 = 0.006 \text{ mol L}^{-1}$ Rate of change of $\Delta[X] = 0.006/6.00 = 0.001 \text{ mol L}^{-1} \text{ s}^{-1}$

Question 28 (5 marks)

28 (a) (2 marks)

Criteria	Marks
• Calculates the correct mass of lead(II) fluoride	2
• Calculates the correct no. of moles of lead(II) fluoride	1

Sample answerLet the solubility of PbF_2 be s .

$$K_{sp} \text{PbF}_2 = 4.0 \times 10^{-8} = [\text{Pb}^{2+}] [\text{F}^-]^2 = (s)(2s)^2 = 4s^3$$

$$4s^3 = 4.0 \times 10^{-8}$$

$$s^3 = 1.0 \times 10^{-8} \text{ mol/L}$$

$$s = 0.00215 \text{ mol/L} = 2.15 \times 10^{-3} \text{ mol/L}$$

Hence mass of lead(II) fluoride that dissolves = $0.00215 \times 245.2 \text{ g} = 0.528 \text{ g} = 0.53 \text{ g}$ (to 2 s.f.)

28 (b) (2 marks)

Criteria	Marks
• Calculates the correct molar solubility of PbF_2 in a solution which contains 0.10 mol of Pb^{2+}	2
• Incorrect answer but evidence of correct working and reasoning	1

Sample answerLet the solubility of PbF_2 in a 0.10 mol/L solution of lead(II) nitrate = s

$$K_{sp} \text{PbF}_2 = [\text{Pb}^{2+}] [\text{F}^-]^2 = (s + 0.10) (2s)^2 = 4.0 \times 10^{-8}$$

Assume s is small by comparison with 0.10,

$$K_{sp} \text{PbF}_2 = [\text{Pb}^{2+}] [\text{F}^-]^2 = (0.10) (4s^2) = 4.0 \times 10^{-8}$$

$$4s^2 = 4.0 \times 10^{-8} / (0.10) = 4.0 \times 10^{-7} \text{ mol/L}$$

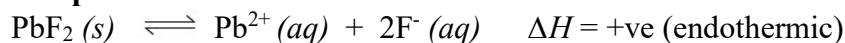
$$s^2 = 1.0 \times 10^{-7}$$

$$s = 0.00032 \text{ mol/L}$$

The solubility of PbF_2 in 0.10 M $\text{Pb}^{2+} = 3.2 \times 10^{-4} \text{ mol/L}$

28 (c) (1 mark)

Criteria	Mark
• Writes a correct equation, showing equilibrium arrows and a correct indication of an endothermic reaction	1

Sample answer

Question 29 (10 marks)

29 (a) (2 marks)

Criteria	Marks
• Writes 2 correct equations	2
• Writes 1 correct equation	1

Sample answer

At pH = 9 $\text{CO}_3^{2-} + \text{H}^+ \rightleftharpoons \text{HCO}_3^-$ (OR $\text{CO}_3^{2-} + \text{H}_3\text{O}^+ \rightleftharpoons \text{HCO}_3^- + \text{H}_2\text{O}$)

At pH = 5 $\text{HCO}_3^- + \text{H}^+ \rightleftharpoons \text{H}_2\text{CO}_3$ (OR $\text{HCO}_3^- + \text{H}_3\text{O}^+ \rightleftharpoons 2\text{H}_2\text{O} + \text{CO}_2$)

29 (b) (2 marks)

Criteria	Marks
• Correctly calculates the number of moles of acid to reach BOTH equivalence points	2
• Correctly calculates the number of moles of acid to reach ONE equivalence point	1

Sample answer

Moles HCl added to reach first equivalence point = $9.63 \times 10^{-3} \times 0.115 \text{ mol}$
 $= 1.107 \times 10^{-3} \text{ mol}$

This has converted carbonate ions into HCO_3^- ions.

Moles HCl added to reach second equivalence point = $34.27 \times 10^{-3} \times 0.115 \text{ mol}$
 $= 3.941 \times 10^{-3} \text{ mol}$

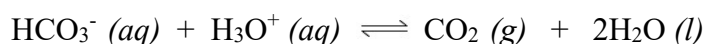
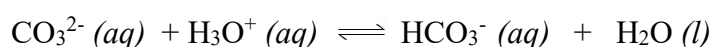
This has converted all HCO_3^- ions into H_2CO_3 (or $\text{CO}_2 + \text{H}_2\text{O}$).

29 (c) (4 marks)

Criteria	Marks
- Correctly determines the number of grams of each of Na₂CO₃ and NaHCO₃ - Correctly determines the mass of KCl in the original sample - Correctly determines the percentages of the 3 salts in the original sample	4
Correctly determines the number of grams of each of Na₂CO₃ and NaHCO₃	3
Correctly determines the number of grams of ONE of Na₂CO₃ and NaHCO₃	2
Shows evidence of correct reasoning but mathematical error(s)	1

Sample answer

The overall reaction occurs in 2 steps, as the carbonate ion is dibasic (can react with 2 hydrogen ions), with each step having an equivalence point.



Moles Na₂CO₃ = no. of moles H⁺ needed to reach the first equivalence point
 = 1.107 x 10⁻³ moles

This has a mass of 1.107 x 10⁻³ x 105.99 g = 0.117 g

Total no. of moles of HCO₃⁻ = total no. of moles of HCl needed to reach the second equivalence point = 3.941 x 10⁻³ mol

Hence total moles HCO₃⁻ = 3.941 x 10⁻³ moles

Of this total of 3.941 x 10⁻³ moles HCO₃⁻, 1.107 x 10⁻³ moles will have been produced from the Na₂CO₃.

Hence (3.941 x 10⁻³) - (1.107 x 10⁻³) = 2.834 x 10⁻³ moles of solid NaHCO₃ must have been part of the original mixture.

This has a mass of 2.834 x 10⁻³ x 84.01 = 0.238 g NaHCO₃

Mass of KCl in original mixture = 0.500 - (0.117 + 0.238) = 0.145 g

Hence percentage of Na₂CO₃ in original solid mixture = 0.117/0.500 x 100 = 23.4%

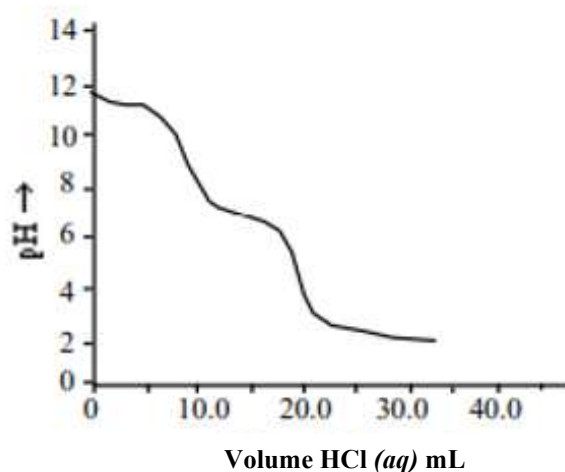
Hence percentage of NaHCO₃ in original solid mixture = 0.238/0.500 x 100 = 47.6%

Hence percentage of KCl in original solid mixture = 0.145/0.500 x 100 = 29.0%

29 (d) (2 marks)

Criteria	Marks
Sketches a graph of the correct shape, showing 2 equivalence points at the correct pH values AND	2
Describes how this differs from the given curve	1

Sample answer



The total volume required to reach the second equivalence point is twice that required to reach the first equivalence point, because the number of moles of HCO_3^- is equal to the number of moles of CO_3^{2-} (all HCO_3^- comes from the original Na_2CO_3). Note: The pH values at the equivalence points are the same as previously.

Question 30 (6 marks)

30 (a) (1 mark)

Criteria	Mark
• Correct answer	1

Sample answer

$$K_{eq} = [\text{NOBr}]^2 / [\text{NO}]^2 [\text{Br}_2] = 1.3 \times 10^{-2}$$

30 (b) (2 marks)

Criteria	Marks
- Correctly determines the quotient (concentration ratio) and identifies that the system is not in equilibrium AND - Predicts that the reverse reaction is favoured to make the system move to equilibrium	2
Correctly determines the quotient (concentration ratio) and identifies that the system is not in equilibrium	1

Sample answer

$$Q = [\text{NOBr}]^2 / [\text{NO}]^2 [\text{Br}_2] = (10.0^2) / (10.0^2 \times 5.0) = 0.20$$

$Q > K_{eq}$, so the system is not in equilibrium. The system shifts to decrease the concentration fraction (at equilibrium $Q = K_{eq}$), and therefore the reverse reaction is favoured.

30 (c) (3 marks)

Criteria	Marks
• THREE correct explanations using collision theory and factors that influence the rate of reaction	3
• TWO correct explanations using collision theory and factors that influence the rate of reaction	2
• ONE correct explanation using collision theory and factors that influence the rate of reaction	1

Sample answer(i) Between t_0 and t_1

The system is at equilibrium, so the rates of the forward and reverse reactions are equal; hence the rate of formation of NOBr is constant. The frequency of successful collisions (between NO and Br₂) is constant.

(ii) At t_1

A volume decrease has the immediate effect of increasing all concentrations, and therefore increasing the frequency (number) of collisions between NO and Br₂, so the rate of formation of NOBr increases.

(iii) Between t_1 and t_2

From t_1 to t_2 , the system is compensating for the pressure increase (due to the volume decrease at t_1) by moving to the side with fewer particles. The associated decrease in [NO] and [Br₂] reduces the frequency of successful collisions and the reaction rate decreases as the system returns to equilibrium. The rate of reaction when the system returns to equilibrium is higher than before the volume decreases because the system only partially compensates for the change when equilibrium is re-established. Therefore, the rate of formation of NOBr is still

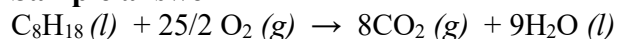
higher than prior to the imposed change because the final concentrations of NO and Br₂ are higher than at the original equilibrium.

Question 31 (5 marks)

31 (a) (1 mark)

Criteria	Mark
• Writes a correctly balanced equation for combustion of octane	1

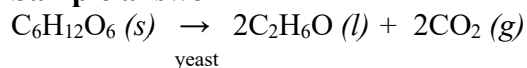
Sample answer



31 (b) (1 mark)

Criteria	Mark
• Writes a correctly balanced equation for the fermentation of glucose to form ethanol	1

Sample answer



31 (c) (3 marks)

Criteria	Marks
• Explains the meaning of “biofuels” • Outlines TWO benefits of the use of a named biofuel • Outlines TWO problems with the use of a named biofuel	3
• TWO of the above	2
• ONE of the above	1

Sample answer (many possible answers)

Biofuels (or biomass fuels) are those biological materials derived from living, or recently living, organisms used to produce energy. Biomass can equally apply to animal- and vegetable-derived material. Examples include ethanol made from sugars or starch by fermentation, firewood, and biodiesel and methane from wastes.

Bioethanol is an alcohol made by fermenting the sugar components of plant materials and it is made mostly from sugar and starch crops.

The advantages of bioethanol include:

- The use of ethanol-blended fuels such as E10 (10% ethanol and 90% gasoline) reduces greenhouse gases by up to 4% over the use of fossil fuels such as petrol.
- Bioethanol burns more cleanly (more complete combustion) than do fossil fuels.

However, bioethanol also has disadvantages for society:

- Production of ethanol requires significant energy and large amounts of land.
- Ethanol has a lower heat of combustion than octane, so less energy is released per gram or per litre.

Section I Mapping Grid

Question	Marks	Content	Syllabus Outcomes
1	1	Mod 6: Quantitative Analysis	12-4, 12-5, 12-13
2	1	Mod 7: Nomenclature	12-5, 12-7, 12-14
3	1	Mod 6: Quantitative Analysis	12-2, 12-3, 12-13
4	1	Mod 5: Factors that Affect Equilibrium	12-5, 12-12
5	1	Mod 8: Analysis of Organic Substances Mod 7: Alcohols, Reactions of Organic Acids and Bases, Nomenclature	12-6, 12-14, 12-15
6	1	Mod 7: Hydrocarbons, Polymers	12-5, 12-14
7	1	Mod 6: Quantitative Analysis	12-6, 12-13
8	1	Mod 5: Factors that Affect Equilibrium	12-4, 12-5, 12-12
9	1	Mod 7: Polymers	12-5, 12-14
10	1	Mod 6: Quantitative Analysis	12-6, 12-13
11	1	Mod 7: Nomenclature, Hydrocarbons	12-7, 12-14
12	1	Mod 5: Factors that Affect Equilibrium	12-5, 12-12
13	1	Mod 8: Analysis of Organic Substances Mod 7: Hydrocarbons	12-5, 12-6, 12-14, 12-15
14	1	Mod 7: Nomenclature	12-7, 12-14
15	1	Mod 5: Dynamic Equilibrium	12-5, 12-12
16	1	Mod 5: Solution Equilibria	12-6, 12-12
17	1	Mod 6: Properties of Acids and Bases	12-7, 12-13
18	1	Mod 8: Analysis of Inorganic Substances	12-5, 12-15
19	1	Mod 8: Analysis of Organic Substances	12-5, 12-15
20	1	Mod 8: Analysis of Organic Substances	12-5, 12-15

Section II Mapping Grid

Question	Marks	Content	Syllabus Outcomes
21 (a)	1	Mod 8: Analysis of Organic Substances, Mod 7: Nomenclature	12–6, 12–7, 12–14, 12–15
21 (b)	1	Mod 7: Reactions of Organic Acids and Bases Mod 8: Analysis of Organic Substances	12–6, 12–7, 12–14, 12–15
21 (c)	1	Mod 8: Analysis of Organic Substances Mod 7: Nomenclature	12–5, 12–14, 12–15
21 (d)	2	Mod 7: Reactions of Organic Acids and Bases	12–5, 12–7, 12–14
21 (e)	1	Mod 7: Reactions of Organic Acids and Bases	12–6, 12–7, 12–14
21 (f)	2	Mod 8: Analysis of Organic Substances Mod 7: Reactions of Organic Acids and Bases	12–5, 12–7, 12–14, 12–15
22 (a)	1	Mod 8: Analysis of Organic Substances	12–6, 12–15
22 (b)	2	Mod 8: Analysis of Organic Substances Mod 6: Quantitative Analysis	12–6, 12–13
22 (c)	1	Mod 8: Analysis of Organic Substances	12–6, 12–15
23 (a)	1	Mod 6: Using Brønsted-Lowry Theory, Quantitative Analysis	12–5, 12–13
23 (b)	1	Mod 6: Using Brønsted-Lowry Theory	12–7, 12–13
23 (c)	1	Mod 6: Quantitative Analysis	12–6, 12–13
23 (d)	2	Mod 6: Quantitative Analysis	12–6, 12–13
23 (e)	2	Mod 6: Quantitative Analysis	12–6, 12–13
23 (f)	2	Mod 5: Factors that Affect Equilibrium Mod 6: Using Brønsted-Lowry Theory	12–4, 12–5, 12–12, 12–13
24	6	Mod 7: Polymers	12–7, 12–14
25 (a)	2	Mod 5: Factors that Affect Equilibrium	12–7, 12–12
25 (b)	3	Mod 5: Factors that Affect Equilibrium, Calculating the Equilibrium Constant	12–6, 12–12
25 (c)	2	Mod 5: Factors that Affect Equilibrium	12–6, 12–12
25 (d)	1	Mod 5: Factors that Affect Equilibrium	12–7, 12–12
26 (a)	3	Mod 8: Analysis of Organic Substances	12–4, 12–6, 12–15
26 (b)	2	Mod 8: Analysis of Organic Substances	12–4, 12–6, 12–7, 12–15
26 (c)	2	Mod 8: Analysis of Organic Substances	12–4, 12–6, 12–7, 12–15
26 (d)	2	Mod 8: Analysis of Organic Substances	12–4, 12–6, 12–7, 12–15
26 (e)	4	Mod 8: Analysis of Organic Substances Mod 7: Reactions of Organic acids and Bases	12–4, 12–6, 12–7, 12–14
27 (a)	1	Mod 8: Analysis of Inorganic Substances	12–4, 12–6, 12–15
27 (b)	1	Mod 8: Analysis of Inorganic Substances	12–4, 12–6, 12–15
27 (c)	2	Mod 8: Analysis of Inorganic Substances	12–4, 12–6, 12–15
27 (d)	2	Mod 8: Analysis of Inorganic Substances	12–4, 12–6, 12–15

Question	Marks	Content	Syllabus Outcomes
28 (a)	2	Mod 5: Solution Equilibria	12–6, 12–12
28 (b)	2	Mod 5: Solution Equilibria	12–6, 12–12
28 (c)	1	Mod 5: Solution Equilibria	12–7, 12–12
29 (a)	2	Mod 6: Quantitative Analysis	12–4, 12–7, 12–13
29 (b)	2	Mod 6: Quantitative Analysis	12–4, 12–6, 12–13
29 (c)	4	Mod 6: Quantitative Analysis	12–4, 12–6, 12–13
29 (d)	2	Mod 6: Quantitative Analysis	12–4, 12–7, 12–13
30 (a)	1	Mod 5: Calculating the Equilibrium Constant	12–4, 12–6, 12–12
30 (b)	2	Mod 5: Calculating the Equilibrium Constant	12–4, 12–6, 12–7, 12–12
30 (c)	3	Mod 5: Factors that Affect Equilibrium	12–4, 12–7, 12–12
31 (a)	1	Mod 7: Hydrocarbons	12–7, 12–14
31 (b)	1	Mod 7: Alcohols	12–7, 12–14
31 (c)	3	Mod 7: Alcohols	12–7, 12–14